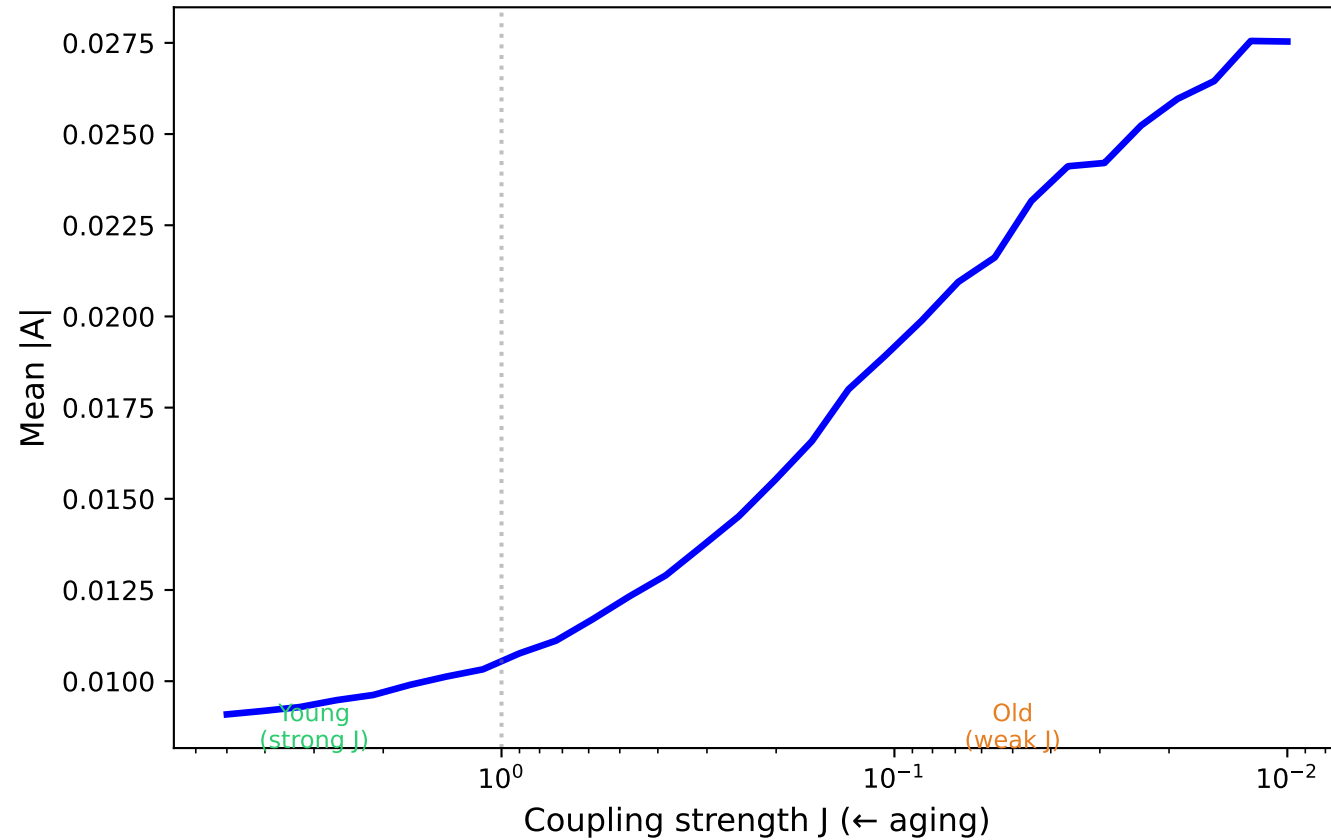
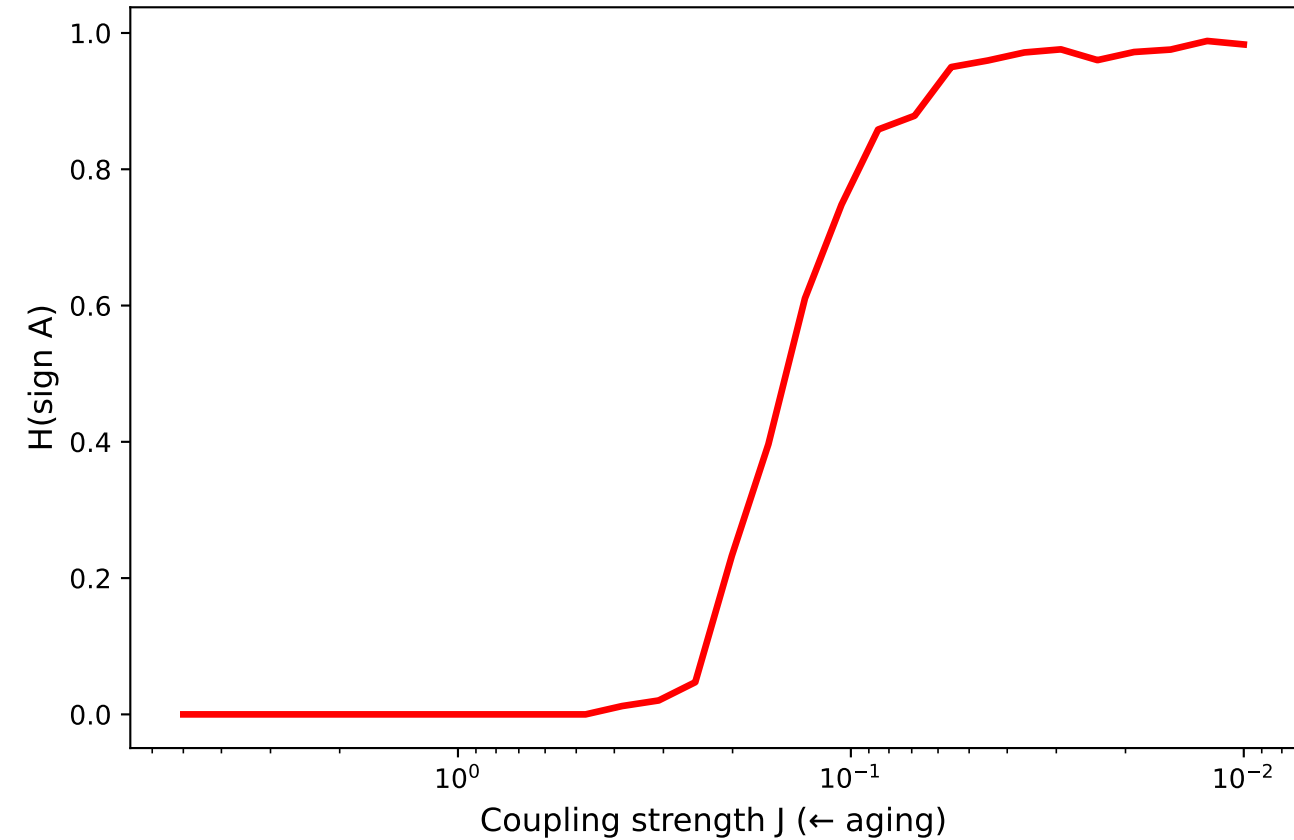


Ising Model of E/I Coarsening: Reduced Coupling → Aging Phenotype

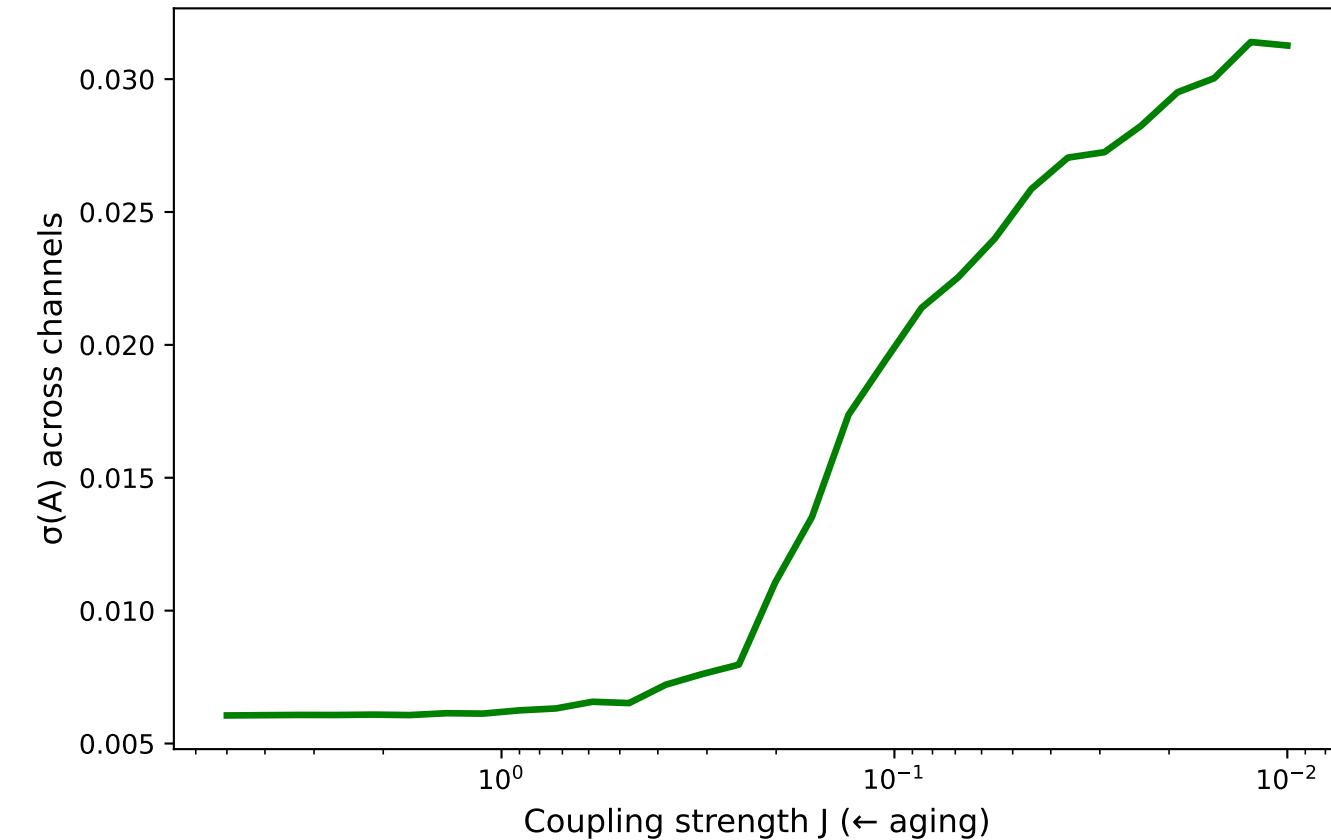
A. $|A|$ increases as coupling weakens



B. Sign entropy decreases as coupling weakens



C. Spatial variability increases as coupling weakens



D. Model ↔ Data comparison

Metric	Model ($J \downarrow$)	Data (age $\uparrow$)	Match?
$ A $	$\uparrow$	$\uparrow$ ($r=+0.45$)	✓
$H(\text{sign})$	$\uparrow$	$\downarrow$ ($r=-0.26$)	✓
$\sigma(A)$	$\uparrow$	$\uparrow$ ($r=+0.47$)	✓

Ising model with decreasing coupling J reproduces all three signatures of age-related EEG coarsening.